

Simulation-Based Conformal Inference

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Abstract

Validating theoretical hypotheses against empirical observations is a major problem in the natural sciences. Given a parametric mathematical model of a phenomenon, such as Newton’s law of gravitation, the task is to find the parameters that best fit the available data. Standard Bayesian inference is computationally expensive for large data sets or multi-dimensional parameters, requires strong assumptions on the process and measurement noise, and is unreliable in the finite-sample regime or when the model is misspecified. Conformal Prediction (CP), a frequentist uncertainty quantification approach, has milder requirements but does not apply straightforwardly to parameter inference as the model parameters (the task labels) are not observable.

As in Simulation-Based Inference, we train a black-box encoder to predict the (unobservable) parameters directly from the observations. The predictor is fitted and calibrated using synthetic (labelled) observations generated by a simulator of the target physical model. Our work is about recovering CP validity on unseen real observations by addressing the unavoidable distributional shifts between synthetic and real data.

1 Introduction

Conformal Prediction (CP) is a well-established uncertainty quantification procedure for evaluating the trustworthiness of machine learning predictors under mild assumptions (Vovk, Gammerman, and Shafer 2005). CP algorithms promote point-like predictions to *prediction regions*, constructed using the predictor’s performance on a *labelled* calibration set via a *conformity score function*.

In real-world parameter inference problems, however, labels (the data-generating parameters) do not exist or are unobservable. Bayesian simulation-based models (Cranmer, Brehmer, and Louppe 2020; Deistler et al. 2025) train a Neural Network to reconstruct the full posterior conditional distribution, which is challenging when the parameter space is multidimensional and requires computationally expensive Markov Chain Monte Carlo (MCMC) sampling to extract the physical parameters from the black-box pseudo-posterior.

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The obtained posterior density is not guaranteed to be correct if the priors are misspecified and the data is finite. Existing CP methods (Hyun and Wang 2025; Baragatti et al. 2026) replace the posterior estimator with a more efficient point regressor but assume the availability of labelled calibration samples, which is only possible if data is simulated. Uncontrolled and unavoidable shifts between simulated and real signals would then *invalidate* the calibrated prediction intervals because calibration and test examples are not exchangeable.

Building on the Feature-CP strategy of (Teng et al. 2022), we design a physics-informed CP pipeline that explicitly addresses the validity issue by estimating the prediction uncertainty in the *latent* space of the physical parameters. A first technical challenge is adapting their definition of *surrogate label* to the physical parameter space, which has fixed (low) dimensionality, specific ranges, interpretability constraints, and does not faithfully represent any observed signal. By letting our surrogate labels depend only on observable quantities, the shift between the synthetic and real data-generating distributions can be estimated and used to restore validity. This is achieved by implicitly defining labels as the deterministic minimizer of a reconstruction cost. The corresponding implicitly defined conformity scores, however, are not observable and must be approximated. We handle the problem by computing prediction regions that combine two simultaneous CP bounds on the estimation accuracy and value of the conformity scores. We then address the distribution shifts between the real and synthetic inputs. Since all inputs are available at test time, the shift is treated as in (Tibshirani et al. 2019) by reweighting the calibration samples with an estimate of the likelihood ratios, $\frac{p_{\text{real}}(y)}{p_{\text{synth}}(y)}$, obtained from a binary discriminator trained on a combination of real and synthetic trajectories.

The scheme is applied to the real-world problem of detecting Earth-like exoplanets, planets orbiting around their stars outside the solar system. Similarity to the Earth is inferred from quantities such as the planet’s orbital period ($\log_{10} P$) and radial-velocity semi-amplitude ($\log_{10} K$), and the orbit’s eccentricity (e), which can only be estimated indirectly by fitting the Kepler equations to sky observations. We use publicly available radial-velocity spectroscopy data from the NASA Exoplanet platform. The application is timely, as the computationally expensive Bayesian approaches used so far struggle with the large-scale data sets produced by today’s

and tomorrow’s state-of-the-art telescopes.

Example. We aim to infer the constant $\bar{\theta}$, appearing in Newton’s differential equations for a pendulum,

$$\frac{d^2 h}{dt^2} = -\bar{\theta} \sin(h), \quad \bar{\theta} = \frac{g}{\ell} \in \Theta \quad (1)$$

assuming g and ℓ cannot be measured directly. The equations do not fully explain an observed trajectory, $\hat{y} = (y_1, \dots, y_T) \in \mathcal{Y} \subseteq \mathbb{R}^T$, $\hat{y} \sim h(\bar{\theta}) + \text{noise}(\bar{\theta})$, where $h(\bar{\theta}) = (h_1, \dots, h_T) \in \mathcal{Y}$ is a noiseless discrete-time integration of (1) and $\text{noise}(\bar{\theta}) \in \mathcal{Y}$ a series of unavoidable and unknown disturbances. Bayesian approaches may capture such estimation uncertainty, but require that (1) is correct and noise is known. Our method works under arbitrary misspecification and without specifying the noise process. We first learn a non-parametric sketch-model, noise , of $\text{noise}(\bar{\theta})$ and, if possible, of the marginal distribution of the data-generating parameters, $p_{\bar{\theta}}$. We then use noise and $p(\bar{\theta})$ to generate a training set of synthetic trajectories $\{y_i \sim h(\bar{\theta}_i) + \text{noise}(\bar{\theta}_i)\}_{i=1}^n$ and use it to learn a *parameter estimator*, $\psi : \mathcal{Y} \rightarrow \Theta$, that recovers $\bar{\theta}$ given y . The estimator is trained by minimizing

$$\begin{aligned} \ell_{\text{train}} &= \ell_{\text{obs}} + \ell_{\text{latent}} \\ \ell_{\text{obs}} &= \|y - h(\psi(y))\|^2, \quad \ell_{\text{latent}} = \|\bar{\theta} - \psi(y)\|^2 \end{aligned} \quad (2)$$

The latent cost is crucial for capturing possibly non-Gaussian deviation from h . Unlike Bayesian inference, ψ is a *point-like* estimator, which does not encode the prediction uncertainty. The CP pipeline aims to *post-hoc* quantify the uncertainty of ψ by measuring the conformity of its prediction using

$$s(y, \theta) = |\psi(y) - \theta| \in \mathbb{R}_+^d \quad (3)$$

where θ is a *computable approximation* of the unobservable latent-space *surrogate label*

$$\theta^* = \arg \min_{\theta' \in \Theta} \|h(\theta') - y\|^2 \quad (4)$$

where the norm is over the observation time steps, $\|z\|^2 = \sum_{t=1}^T z_t^2$. Crucially for the validity of the CP pipeline, θ^* is *conditionally independent of $\bar{\theta}$ given y* . The computable approximation of θ^* ,

$$\theta = \theta^{(p)}, \quad \theta^{(r+1)} = \theta^{(r)} - \eta \nabla \ell(\theta^{(r)}) \quad (5)$$

where $\theta^{(0)} = \bar{\theta}$, $\eta > 0$ guarantees $\ell(\theta^{(r+1)}) < \ell(\theta^{(r)})$, $r = 0, \dots, p-1$, and $\ell(\bar{\theta}) = \|y - h(\bar{\theta})\|^2$, is also conditionally independent of $\bar{\theta}$ given y .¹ Then $p(s, \theta, y) = p(s|\theta, y)p(\theta|y)p(y)$ ² and the distribution shift can be treated

¹We require the process and measurement noise to be such that $\bar{\theta}$ is in the basin of attraction of θ^* and there exists p for which a θ is independent of $\bar{\theta}$.

$$\begin{aligned} p(s, \theta, y) &= \sum_{\bar{\theta}} p(s, \theta, y|\bar{\theta})p(\bar{\theta}) \\ &= \sum_{\bar{\theta}} p(s|\theta, y)p(\theta|y)p(y|\bar{\theta})p(\bar{\theta}) = p(s|\theta, y)p(\theta|y)p(y) \end{aligned} \quad (6)$$

using the covariate-shift approach of (Tibshirani et al. 2019). Given a held-out calibration set of synthetic trajectories, $\{y_i \sim h(\bar{\theta}_i) + \text{noise}(\bar{\theta}_i)\}_{i=n+1}^{2n}$, we construct the CP regions by estimating the empirical quantiles of $F_{\text{real}}(s) \approx \sum_{i=n+1}^{2n} w_i \mathbf{1}(s = s(y_i))$, where w_i are proportional to the likelihood ratios $\frac{p_{\text{real}}(y)}{p_{\text{synth}}(y)}$.

Contribution To the best of our knowledge, this is the first CP scheme for simulation-based inference that does not require reference labels. Tabulated values are used to fit the prior marginal distribution of the target parameters, which is not essential to the method and can be replaced by ranges derived from more general prior knowledge. The connection between the simulation-based framework and latent-space uncertainty estimation is also new. We are unaware of existing CP methods with labels that are potentially observable in both the training and test data sets but can be safely approximated only on the training set. Technical contributions include i) a general scheme for calibrating CP with *approximate* labels, ii) a pseudo-label approach for uncertainty quantification in a physically constrained latent space without minimizing distribution-based Bayesian-like objectives³, and iii) a CP method for simulation-based inference that is robust to model misspecification in the simulator and noise model with finite-sample guarantees⁴.

From an astrophysical perspective, ours is the first non-Bayesian approach for identifying Earth-like exoplanets with quantified uncertainty. The computational savings allow extending exoplanet searches to the millions of data points generated by present-day telescopes. The proposed pipeline may be used to combine Radial Velocity (RV) and Transit exoplanet detection methods with minor changes.

Related work

Parameter estimation of physical models Our approach is similar to Simulation-Based Inference, a popular Bayesian approach for parameter estimation in Physics (Cranmer, Brehmer, and Louppe 2020; Deistler et al. 2025) where the ML model approximates the full posterior distribution (Lueckmann et al. 2017; Radev et al. 2020) and is used to reduce the computational complexity of full likelihood estimations (Greenberg, Nonnenmacher, and Macke 2019; Gonçalves et al. 2020). Similar to (Hyun and Wang 2025; Baragatti et al. 2026), our ML model is a point predictor, with Bayesian inference replaced by a CP.⁵ Training a point-like estimator of the parameter is easier than training a model for the conditional posterior density. Using CP, the uncertainty estimation does not depend on the modelling assumptions, that is, the correctness of the simulator.

CP for system identification. CP has been used to quantify system identification uncertainties in (Teng et al. 2022) and (Msaddi and Kumbasar 2025), which focus on bounding observable outputs instead of the identified parameters.

³As in the MC Dropout estimation of (Baragatti et al. 2026)

⁴Unlike (Hyun and Wang 2025), which is valid only asymptotically

⁵See section 1 for a technical comparison with these methods.

Computing the prediction regions in the parameter space is more challenging. Our simulation-based approach differs from the masking techniques used to train Foundation Models (Wiesner et al. 2025; Parker et al. 2026; Tame-Narvaez, Čiprijanović, and Trivedi 2026). In that framework, uncertainty quantification is fully supervised.

CP with pseudo or uncertain labels. We are not aware of any existing CP approach for *non-computable* labels, where the targets are deterministic given an observation but known only approximately. A few recent works have addressed the problem of uncertain labelling. Stutz et al. (2023) uses a multiple-expert scenario, is restricted to classification tasks, and requires sampling from the label empirical distribution. Cabezas et al. (2025) models the epistemic uncertainty of the labels with a learned mixture of prior distributions. Similar to us, Bashari et al. (2026) improves CP calibration using synthetic scores via a map between their quantiles and those of the true data but requires a possibly small data set of true and known labels. The method of (Bashari et al. 2023) and Prediction Powered Inference algorithms (Angelopoulos et al. 2023) are calibrated using predicted labels but assume unlabelled data are drawn from the same distribution as the labelled data and require a set of unambiguous real labels. Closer to our framework, Teng et al. (2022) uses pseudo-labels defined in a hidden space. See below for a detailed comparison.

Exoplanets detection. We are not aware of any non-Bayesian exoplanet discovery scheme. The PYANETI suite of (Barragán, Gandolfi, and Antoniciello 2019) is the reference Bayesian approach for estimating the parameters of planetary systems from both RV and Transit data. More efficient, gradient-based optimization strategies for it are implemented in the `exoplanet` Python library (Eastman, Gaudi, and Agol 2013). See (Wright and Gaudi 2012) or (Hatzes 2019) for an introduction to these approaches and (Fischer et al. 2015) for a review of existing work. A guided introduction to Kepler dynamics can be found in (Murray and Correia 2010). An inference methodology similar to ours was proposed in (Gaskin, Pavliotis, and Girolami 2023; Gaskin et al. 2024) in the multi-agent systems domain.

Comparison to (Hyun and Wang 2025) and (Baragatti et al. 2026). Two existing works propose using NNs and CP for parameter identification. Hyun and Wang (2025) focuses on diffusion processes and, unlike here, calibrates the CP algorithm using the data-generating parameters. This implicitly requires the model to be correctly specified⁶ and may compromise the validity of prediction regions when it is not. The authors address the distribution shift problem by making the algorithm input-adaptive as in (Guan 2019), but it is valid only in the limit of infinitely many calibration samples. The technique of (Baragatti et al. 2026) is straightforward (no simulation involved) but requires a set of fully trusted ground-truth parameters, which are unavailable in our setup. The authors focus on multi-dimensional prediction sets using a (possibly) more efficient strategy similar to (Messoudi,

Destercke, and Rousseau 2021).⁷

Comparison with (Teng et al. 2022). To the best of our knowledge, surrogate labels were first used in (Teng et al. 2022) to perform CP in the latent space. As the latent space is essentially unconstrained, surrogate labels are defined as $\theta \in \{\Theta : h(\theta) = y\}$, which does not apply to our framework, where the physical constraints may imply $\{\Theta : h(\theta) = y\} = \emptyset$. Similar to us, Teng et al. (2022) uses a minimization step to compute the conformity score, but focuses on improving the efficiency of the CP algorithm in the observable space. In system identification tasks, a conformity score defined in the observable space as in (Teng et al. 2022) would require practically challenging steps to extract the prediction regions in the parameter space.

2 Methods

Definitions and data As in the example above, we use θ , θ^* , and $\bar{\theta}$ for the data-generating parameters, the surrogate labels, and their GD-inferred approximations, $h(\bar{\theta})$ for a noiseless simulation of the target physical model with parameter $\bar{\theta}$, $\hat{y} \sim h(\bar{\theta}) + \text{noise}(\bar{\theta})$, where noise is an unknown misspecification error, for real trajectories, $\hat{y} \sim p_{\text{real}}$, and $y \sim h(\bar{\theta}) + \text{noise}(\bar{\theta})$, where noise is a fitted noise model, for $y \sim p_{\text{synth}}$. For simplicity, we assume that all elements of $\mathcal{Y}$ consist of T time steps and include the integration initial condition as parameters. We split the available real time series and the corresponding tabulated parameters $\{\bar{\theta}_j, \hat{y}_j\}_{j=1}^{3n}$ into *training*, *validation*, and *test* sets, D_{real} , D_{val} , and D_{test} . D_{real} is used for learning a proxy of the parameter *marginal distribution*, $p_{\bar{\theta}}$, given by d independent histogram representations of $\{\theta_{jk} \in D_{\text{real}}\}_{j=1}^n$, $k = 1, \dots, d$, and a non-parametric noise model $\text{noise} \approx \hat{\text{noise}}$ defined as

$$\text{noise}(\bar{\theta}) \sim \hat{y} - h(\bar{\theta}) | \bar{\theta} \quad (7)$$

D_{val} and D_{test} are held out to tune the hyperparameters of the CP model through a cross-validation loop, and to validate the empirical validity and efficiency of the final model. We generate $2n$ synthetic trajectories

$$y_i \sim h(\bar{\theta}_i) + \text{noise}(\bar{\theta}_i), \quad \bar{\theta}_i \sim p_{\bar{\theta}} \quad (8)$$

$i = 1, \dots, 2n$, compute the corresponding GD-inferred labels, θ_i defined in (5), and form a *proper training and calibration* data sets, $D_{\text{train}} = \{(y_i, \theta_i)\}_{i=1}^n$ and $D_{\text{cal}} = \{(y_i, \theta_i)\}_{i=n+1}^{2n}$. Using $\bar{\theta}$ as a proxy for θ^* is equivalent to setting $p = 0$ in (5). In this case, $p(s, \theta, y) = \sum_{\bar{\theta}} p(s | \bar{\theta}, y) p(\theta | \bar{\theta}, y) p(y | \bar{\theta}) p(\bar{\theta})$, and the approximation $p(s | \theta, y) \approx p(s | y)$ is not justified. In Assumption 3.1 below, we require θ to be in the *basin of attraction* of θ^* , which makes the approximation more reasonable if $\theta = \theta^{(p)}$ with $p \gg 1$. The conditional independence of s from $\bar{\theta}$ given y is what allows us to mitigate the distribution shift between synthetic and real data by reweighting the samples with $w(y) \propto \frac{p_{\text{real}}(y)}{p_{\text{synth}}(y)}$.

⁶See Equation 1 in (Hyun and Wang 2025).

⁷Any advanced multiple-output CP approach can be added on top of our scheme with minor changes.

The regression model. D_{train} is used to learn an encoder-decoder regression model,

$$\psi = g \circ f : \mathcal{Y} \rightarrow \Theta \quad f : \mathcal{Y} \rightarrow \mathbb{R}^l, \quad g : \mathbb{R}^l \rightarrow \Theta \quad (9)$$

where f is a non-parametric feature extractor and g a trainable regression head by minimizing (2).

Uncertainty estimation. We use D_{cal} to *calibrate* a multiple-output CP algorithm based on ψ (Vovk, Gammerman, and Shafer 2005; Angelopoulos et al. 2023). The CP algorithm evaluates the *conformity score* defined in (3), $s(y)$, on the calibration samples to quantify the prediction accuracy of ψ and *predict its uncertainty* on a new unseen instance. For any real test trajectory, $\hat{y} \in D_{test}$, the prediction region is the hyper-rectangle defined by

$$\Gamma(\hat{y}) = [\psi(\hat{y}) - q, \psi(\hat{y}) + q] \subseteq \mathbb{R}^d \quad (10)$$

$$q_k = \text{quantile} \left(1 - \alpha'; \sum_{i=n+1}^{2n} \tilde{w}_i \delta_{s_{ik} + \Delta s_k} + \tilde{w}_{test} \delta_{\infty} \right)$$

$$\Delta s_k = \text{quantile} \left(1 - \beta'; \sum_{i=n+1}^{2n} \tilde{w}_i \delta_{r_{ik}} + \tilde{w}_{test} \delta_{\infty} \right)$$

$$\tilde{w}_i = \frac{w_i}{\sum_{i=n+1}^{2n} \tilde{w}_i + w_{test}}, \quad w_i = \frac{p_{real}(y_i)}{p_{synth}(y_i)}$$

$$\tilde{w}_{test} = \frac{w_{test}}{\sum_{i=n+1}^{2n} \tilde{w}_i + w_{test}}, \quad w_{test} = \frac{p_{real}(\hat{y})}{p_{synth}(\hat{y})}$$

where $\alpha' = \frac{\alpha}{d}, \beta' = \frac{\beta}{d}$, with $1 - \alpha, 1 - \beta \in (0, 1)$ being user-defined confidence levels. In practice, we use approximate likelihood ratios,

$$w_i \approx \frac{\phi(y_i)}{1 - \phi(y_i)}, \quad w_{test} \approx \frac{\phi(\hat{y})}{1 - \phi(\hat{y})} \quad (11)$$

where $\phi(\tilde{y}) \approx \text{Prob}(\tilde{y} \in D_{real} | \tilde{y})$, is a binary classifier trained to discriminate between synthetic and real trajectories, $\tilde{y}_i = y_i \in D_{train}$, with label 0, and $\tilde{y}_i = \hat{y}_i \in D_{real}$ with label 1.⁸ According to (Tibshirani et al. 2019), the likelihood ratios ensure validity under the covariate shift, $p_{real}(y) \neq p_{synth}(y)$. See (Tibshirani et al. 2019) for more details about this sample-reweighting technique. Assuming $p(\theta, \bar{\theta}, y) = p(\theta|y)p(y|\bar{\theta})p(\bar{\theta})$ does not make the problem trivial because $\theta = \theta^{(p)}$ in (5) cannot be obtained without a good initialization, $\theta^0 = \bar{\theta}$. The *corrected confidence levels*, α' and β' ensure the simultaneous validity of the d dimensional prediction interval, $\Gamma(\hat{y}) = \Gamma_1(\hat{y}) \otimes \dots \otimes \Gamma_d(\hat{y})$, $k = 1, \dots, d$, via the Bonferroni inequality. Unlike usual CP, q_k is the quantile of the *inflated conformity scores*, $s_{ik} + \Delta s_k$ to guarantee the validity of the prediction region

$$\text{Prob}(\theta_{test}^* \in \Gamma(\hat{y})) \geq 1 - \alpha - \beta \quad (12)$$

where $\theta_{test}^* = \arg \min_{\theta'} \|\hat{y} - h(\theta')\|^2$. Finally, the encoder-decoder approach allows profiling the uncertainty estimation

⁸When a different number of synthetic and calibration samples are available, an extra class-imbalance factor is added to $\frac{\phi(y_i)}{1 - \phi(y_i)}$.

without fitting additional conditional models. For example, we can replace $s(y)$ defined in (3) with

$$\tilde{s}_k(y) = \frac{|\psi_k(y) - \theta_k|}{a + b \delta_k(y) + (1 - b) \delta_y(y)} \quad (13)$$

$$\delta(y) = |\psi(h(\psi(y))) - \psi(y)| \in \mathbb{R}_+^d$$

$$\delta_y(y) = \|y - h(\psi(y))\| \in \mathbb{R}_+$$

where $a > 0$ and $b \in (0, 1)$ are hyperparameters to be fixed by cross-validating the size and empirical coverage of the corresponding *profiled prediction regions*. The profiled prediction region, $\tilde{\Gamma}(\hat{y})$ is defined as in (10) with q replaced by

$$\tilde{q} = q \circ (a + b \delta(\hat{y}) + (1 - b) \delta_y(\hat{y})) \quad (14)$$

Compared to (10), the width of $\tilde{\Gamma}$ depends on $\hat{y}$, which improves its adaptivity if data are heteroskedastic (Papadopoulos, Gammerman, and Vovk 2008; Bellotti 2021; Colombo 2023).

3 Theory

Assumptions We make two mild assumptions on the discrepancy between the target physical model, h , and its noisy realizations, y and $\hat{y}$. We require noise or noise to be such that an approximation of the surrogate label, θ^* defined in (4), can be obtained using a GD algorithm *provided we are given a good initialization*. As such an initialization is unavailable at test time, the iterative algorithm is unusable or untrustworthy, and the prediction and uncertainty estimation tasks are meaningful.

Assumption 3.1 (Generating function). *Observed or generated trajectory, $y, \hat{y} \in \mathcal{Y}$, can be written as in (8) for some data-generating parameter, $\bar{\theta}$. The unknown noise model (7) is such that $\bar{\theta}$ belongs to the basin of attraction of the unavailable surrogate label (4).*

Assumption 3.2 (GD trajectories p). *Under Assumption 3.1, there exists p_{min} such that the GD-inferred label, $\theta = \theta^{(p)}$ defined in (5) is independent of $\bar{\theta}$ if $p > p_{min}$.*

Assumption 3.1 does not explicitly constrain p_{real} , as the unknown noise model, $\text{noise}(\bar{\theta})$, is not required to be of a specific form, such as, for example, a white or coloured Gaussian. Naturally, Assumption 3.1 is expected to hold if the target physical model is nearly correct. Formulating constraints on the energy landscape of the loss defined in (4) is also a clean way to handle possible parameter degeneracies. Assumption 3.2 requires all GD trajectories started at different points in the basin of attraction of θ^* to become nearly indistinguishable near θ^* .

Validity. Under Assumptions 3.1 and 3.2, we first bound the error of calibrating the CP algorithm with an approximate conformity score, s in (3) instead of $s^* = \|\psi(y) - \theta^*\|$, and then account for the non-exchangeability of $\hat{y} \sim p_{real}$ and $y_i \sim p_{synth}$. In the remaining of this section, we let $D_{cal}, \theta, \theta^*, \bar{\theta}_i, s, s^*$, and $\tilde{w}_i$ be the calibration set defined in Section 2, the GD-inferred label (5), the data-generating parameter introduced in (8), the surrogate label (4), the approximate

conformity score (3), its exact counterpart $s^* = \|\psi(y) - \theta^*\|$, and the likelihood ratio-weights defined in (10). Under Assumptions 3.1 and 3.2, we let $p > p_{min}$ be the number of GD steps in (5). To handle approximate labels, we find a conformal upper bound for s^* in terms of s and the quantile of $\Delta s = |s - s^*|$.

Lemma 3.3 (CP bound on Δs). *If $y_i, i = n+1, \dots, 2n$ and $\hat{y}$ are exchangeable,*

$$\text{Prob}(s_{test\ k}^* \leq s_{test\ k} + \Delta s_k) \geq 1 - \beta \quad (15)$$

$$\Delta s_k = \text{quantile} \left(1 - \beta; \frac{1}{n+1} \sum_{i=n+1}^{2n} \delta_{r_{ik}} + \delta_\infty \right)$$

$$r_{ik} = |s_{ik} - |\psi_k(y_i) - \lim_{p \rightarrow \infty} \theta_{ik}^{(p)}||$$

for any $\beta \in (0, 1)$.

Proof of Lemma 3.3. Under Assumption 3.1, $\theta^* = \lim_{p \rightarrow \infty} \theta^{(p)}$ because $\bar{\theta}$ belongs to the basin of attraction of θ^* . From the definition of s^* , we have $s_k^*(y) = |\psi(y) - \theta^*| = |\psi(y) - \lim_{p \rightarrow \infty} \theta^{(p)}|$. The statement follows from Lemma 1 of (Tibshirani et al. 2019) with conformity scores $r_k(y) = |s_k(y) - |\psi_k(y) - \lim_{p \rightarrow \infty} \theta^{(p)}||$. $\square$

Using the surrogate labels in (4), the distribution shift between synthetic and real trajectories is *observable* and can be addressed as in (Tibshirani et al. 2019).⁹ The observability of the shift follows from Assumption 3.2, which disentangles the GD-inferred label and the data-generating parameter, $\theta^{(0)} = \bar{\theta}$, when $p > p_{min}$.

Lemma 3.4 (covariate shift, surrogate labels).

$$\text{Prob}(s_k^*(\hat{y}) \leq q_k) \geq 1 - \alpha \quad (16)$$

$$q_k = \text{quantile} \left(1 - \alpha; \sum_{i=n+1}^{2n} \tilde{w}_i \delta_{s_{ik}^*} + \tilde{w}_{test} \delta_\infty \right) \quad (17)$$

for any $\alpha \in (0, 1)$.

Proof of Lemma 3.4. The conditional distribution of θ^* given y is the same for calibration and test objects because the minimization in (4) is deterministic. As $p > p_{min}$, Assumption 3.2 implies

$$p(s, \theta, y) = \sum_{\bar{\theta}} p(s|\theta, y) p(\theta|y, \bar{\theta}) p(y|\bar{\theta}) p(\bar{\theta}) \quad (18)$$

$$= \sum_{\bar{\theta}} p(s|\theta, y) p(\theta|y) p(y|\bar{\theta}) p(\bar{\theta}) = p(s|\theta, y) p(\theta|y) p(y) \quad (19)$$

As $p(\theta|y)$ does not shift at test time if the optimization algorithm remains the same, the distribution shift is a covariate shift. The validity of the bound follows from Corollary 1 of (Tibshirani et al. 2019). $\square$

The gap between exact and approximate scores, s^* and s , which can also be affected by distribution shift, is similarly adjusted using the reweighted quantiles defined in (10).

⁹See also (Yang, Kuchibhotla, and Tchetgen Tchetgen 2024) for a synthetic description of the reweighting strategy.

Lemma 3.5 (covariate shift, delta s).

$$\text{Prob}(s_{test\ k}^* \leq s_{test\ k} + \Delta s_k) \geq 1 - \beta \quad (20)$$

$$\Delta s_k = \text{quantile} \left(1 - \beta; \sum_{i=n+1}^{2n} \tilde{w}_i \delta_{r_{ik}} + \tilde{w}_{test} \delta_\infty \right) \quad (21)$$

where r_i are defined in (15) and $\beta \in (0, 1)$.

Proof of Lemma 3.5. The proof is analogous to Lemma 3.4 after replacing the conformity scores with r_{ik} defined in (15). As $r_k = r_k(\theta, \theta^*, y)$ and θ^* is a deterministic function of y , the statement follows from (18) and Corollary 1 of (Tibshirani et al. 2019). $\square$

Lemmas 3.5 and 3.4 can be combined to prove the validity of the proposed end-to-end estimation approach.

Theorem 3.6. *For any $\alpha, \beta \in (0, 1)$, let $\Gamma(\hat{y})$ be as in (10) with $\alpha' = \frac{\alpha}{d}$, $\beta' = \frac{\beta}{d}$. Then $\Gamma(\hat{y})$ obeys (12).*

Proof of Theorem 3.6. The theorem follows from Lemmas 3.4 and 3.5. For each k , let $A_k = s_k(\hat{y}) + \Delta s_k \leq q_k$ and $B_k = s_k^*(\hat{y}) - s_k(\hat{y}) \leq \Delta s_k$. Then, Lemmas 3.4 and 3.5 imply

$$\text{Prob}(s_k^*(\hat{y}) \leq q_k) \geq \text{Prob}(A_k|B_k) \text{Prob}(B_k) \quad (22)$$

$$\geq (1 - \alpha')(1 - \beta') \geq 1 - \alpha' - \beta' \quad (23)$$

when q_k and Δ_k are as in (10). Bonferroni's inequality guarantees that the d events hold simultaneously with probability greater than $1 - \alpha - \beta$. $\square$

Bonferroni multiple-hypothesis testing guarantees that the d -dimensional prediction regions are valid but may be inefficient when the target variables are strongly correlated. Tighter uncertainty regions may be obtained by using error-reweighted conformity scores (13) (Papadopoulos, Gammernan, and Vovk 2008) or advanced techniques for multiple-output conformal regression (Messoudi, Destercke, and Rousseau 2021). The quantile adjustment defined in Lemma 3.3 recovers validity when only approximate labels are available for calibration. In practice, the approximation error, Δs , depends on the number of GD steps, p , used in (5). If p is large enough, Δs can be expected to be small. According to Theorem 3.6, the prediction intervals obtained by setting $\Delta_s = 0$ would be valid at the lower confidence level $1 - \alpha - \text{gap}$, where

$$\text{gap}(q) = \left| \frac{1}{n} \sum_{i=n+1}^{2n} w_i \mathbf{1}(q - \Delta s_k \leq s_{ik} \leq q) \right| \quad (24)$$

In the experiments, where Assumption 3.1 is not guaranteed to hold, we set $\Delta s = 0$ and evaluate the associated validity gaps empirically.

4 Experiments

Synthetic Experiment: Forced Pendulum Oscillations

As a controlled test, we run a synthetic experiment using realistic simulations. We augment the pendulum in (1) with a nonlinear forcing term, $\ddot{y}(t) = -\theta \sin y(t) + \lambda \cos(1.7t)y(t)(1 - y(t)^2)$, and infer the two latent parameters (θ, λ) , where $\theta = g/\ell$. The estimator is trained on

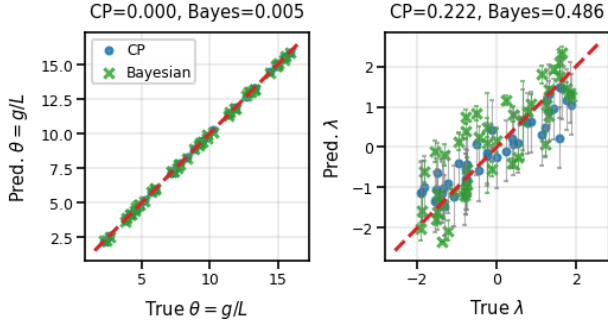


Figure 1: Forced-pendulum parameter identification on held-out realistic trajectories. Error bars show conformal intervals and Bayesian one-standard-deviation intervals; hidden true values are used only for evaluation.

labelled synthetic trajectories from the ideal simulator, and the conformal box is calibrated on a separate synthetic calibration set. Held-out “realistic” trajectories include *Gaussian measurement noise* and occasional state perturbations, which mimic the real-world misspecification of the physical model. Labels are hidden during training and calibration and used only for evaluation. We compare with Bayesian one-standard-deviation intervals, with the posterior evaluated by exhaustive search on the 2-dimensional parameter space. We set the conformal target to joint confidence $1 - \alpha = 0.68$, using coordinate confidence $1 - (1 - 0.68)/2 = 0.84$.

Figure 1 shows that θ is easy while the forcing coefficient λ is the informative failure mode. The conformal point estimator has lower MSE on both coordinates, 4.7×10^{-4} for θ and 0.222 for λ , compared with 5.2×10^{-3} and 0.486 for the Bayesian posterior mean. At nominal joint confidence 0.68, the conformal box contains the hidden pair for 35/45 held-out trajectories, while the Bayesian one-standard-deviation rectangle contains it for 2/45.

Real-world experiment: Keplerian parameters from radial velocity

We apply the scheme to radial-velocity time series from the NASA Exoplanet Archive, inferring $\theta = (\log_{10} P, \log_{10} K, e)$ — orbital period, semi-amplitude and eccentricity — from 1,071 curves across 411 host stars. Splitting by host and keeping the best-sampled series per star leaves 34 independent test systems. The simulator h solves Kepler’s equation; the noise model in (7) is a sparse variational Gaussian process fitted to training-curve residuals, with observing cadences bootstrapped from real curves. Full detail is in the supplementary material.

A feed-forward ψ cannot on its own reach the precision a trajectory requires: the observing baseline spans many orbits, and phase coherence across it constrains the period to better than a part in 10^3 . We therefore report the refined predictor $\psi'(y) = \text{GD}_K(\psi(y))$, i.e. K descent steps on $\ell(\theta) = \|h(\theta) - y\|^2$ started from the learned estimate and restarted from the strongest periodogram peaks. Fitting ψ' requires no label and is applied identically to the calibration

Nominal	Region	$\log_{10} P$	$\log_{10} K$	e
0.90	box	3.326	1.964	1.534
	union	2.527	1.135	1.392
0.68	box	1.776	0.986	1.160
	union	1.016	0.491	1.087

Table 1: Total reported measure of the single box against the union of the per-candidate intervals. The refined predictor produces one fit per periodogram restart; the box keeps the lowest-loss fit and reports one interval around it, whereas the union scores against the *nearest* fit and reports the merged union of one interval per candidate. Both are calibrated identically. Measures are over the 34 independent hosts, likelihood-ratio weighted, and are twice the half-widths quoted in the text. Joint coverage is 1.000 for both shapes at nominal 0.90, and 0.882 (box) against 0.853 (union) at 0.68.

and test sets, so CP validity follows from Theorem 3.6 unchanged. Figures 3 and 4 show that ψ' matches or beats the published Bayesian fit on 26 of the 51 held-out curves (the 34 hosts before per-host de-duplication), against 2% before refinement; its median reconstruction error is 0.176, against 0.090 for the catalogue and 0.072 for the surrogate label itself.

Figure 2 shows the prediction regions for the ten most Earth-like held-out planets. Coverage meets or exceeds nominal — 34/34 at 0.90 and 0.88 at 0.68, with no unbounded interval and an effective calibration sample size of 262.6 of 400 — as expected, since Theorem 3.6 is a lower bound. Validity there is claimed with respect to the surrogate label θ^* , whereas the coverage reported here is measured against the tabulated catalogue values, the only reference available on real curves; the two targets agree closely on this corpus (median MSE 0.072 against 0.090). Median half-widths at nominal 0.90 are 1.663 and 0.982 dex in $\log_{10} P$ and $\log_{10} K$, and 0.767 in e .

As the residual period ambiguity is discrete — aliases rather than a continuum — we also report the union of the per-candidate intervals, one per periodogram restart, which is calibrated identically and changes only the shape of the region. At nominal 0.68 this reduces the reported measure by 43% in $\log_{10} P$ and 50% in $\log_{10} K$, at joint coverage 0.85 against 0.88 for the single box (Table 1). The reduction is concentrated in the period, the only genuinely multimodal coordinate: in e the candidates almost coincide and the union buys 6%.

The comparison against a conventional baseline is one-sided by construction: calibrating on $\bar{\theta}$ — ordinary split CP, without surrogate labels or reweighting — reaches 0.932 against the surrogate’s 0.915 on synthetic curves, but cannot be carried out at all on real systems, where $\bar{\theta}$ does not exist. The procedure is also markedly conservative on real data, at 30 of 34 even at nominal 0.68; with 34 systems a single host moves the figure by three points, so we rest no claim on comparing real-data coverage between variants.

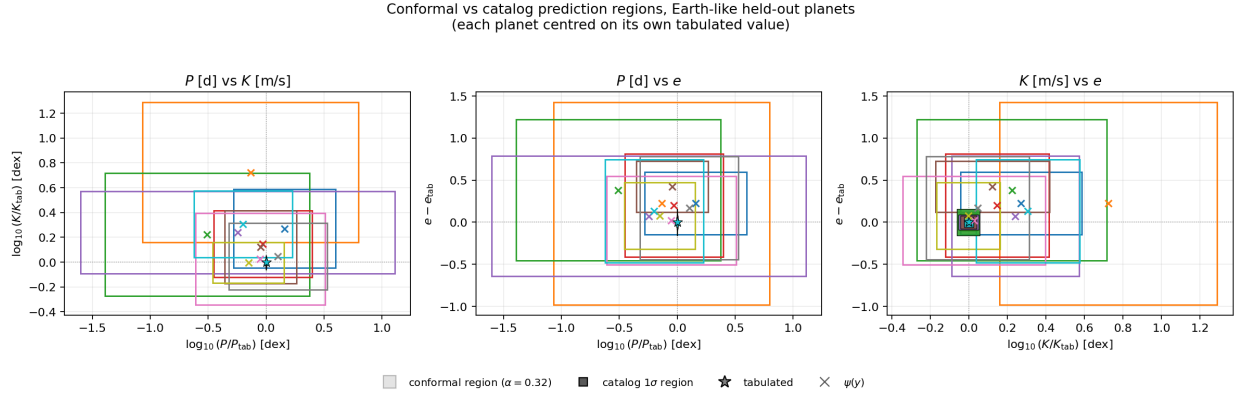


Figure 2: Conformal regions at nominal 0.68 against the catalogue’s 1σ credible regions, for the ten most Earth-like held-out systems, in three two-dimensional projections. Each planet is centred on its own tabulated value, so the origin is the catalogue answer and the dark box is its 1σ region. Crosses mark the point prediction $\psi'(y)$. The conformal regions are rectangles by construction, being products of per-coordinate intervals; they cover the tabulated value jointly on 7 of the 10 systems, above nominal.

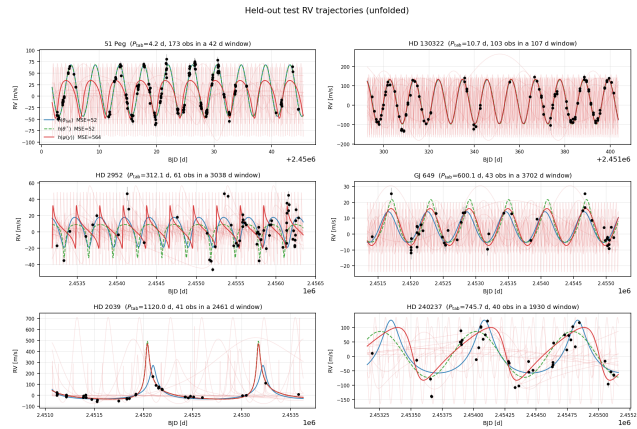


Figure 3: Held-out RV trajectories: noiseless curves at the tabulated parameters, at the surrogate label θ^* and at the refined predicted parameters; faint curves sampled from the conformal region; and the observations with their measurement uncertainties. The legend, in the first panel, gives the MSE of each curve against the data.

5 Conclusion

This work proposes and tests a new end-to-end CP pipeline for identifying the parameters of dynamical systems from time-series data and quantifying the uncertainty of the identification. Assuming a simulator of the target system is available, we calibrate an ML point predictor on simulated data using CP, and then address the unavoidable distribution shift between synthetic and real data so that the prediction regions remain valid on real inputs. On radial-velocity observations of exoplanet hosts, the regions attain nominal coverage with no unbounded interval, and reporting a union of per-candidate intervals rather than a single box halves the reported measure in $\log_{10} K$. Coverage on real data is measured against tabulated catalogue values rather than the surrogate

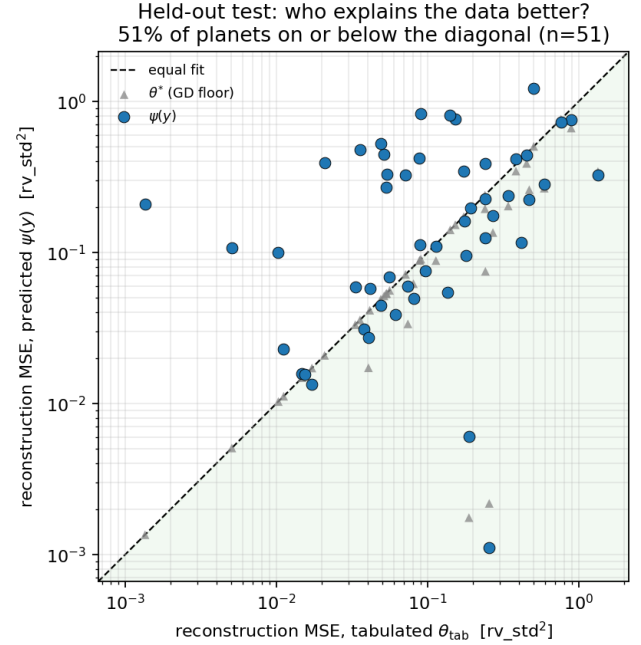


Figure 4: Reconstruction MSE of the predicted parameters against the tabulated ones, per held-out system. The surrogate-label floor θ^* lies below the tabulated values on most systems.

labels the theory bounds, and we leave a comparison against simulation-based posterior estimators to future work.

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